

Lesson 3: Remote Repositories, Pull Requests & GitHub Collaboration

In Lesson 2, you mastered branching and merging on your local computer. Now, we expand into true team collaboration by connecting your local Git repository to a remote server using GitHub.

1. Local vs. Remote Repositories

Up until now, all commits lived strictly on your hard drive. A Remote Repository is a version of your project hosted on the cloud (e.g., GitHub, GitLab, Bitbucket).

- Local Repo: Lives on your machine; fast, offline, and personal.
- Remote Repo: Lives on a cloud server; serves as the central "single source of truth" for the whole team.

[Your Computer] [GitHub Cloud]

Local Repo ----- (git push) ----> Remote Repo ('origin')

<--- (git fetch / git pull) -----

2. Connecting Local Git to GitHub

To link a local project to GitHub, you register a remote reference (conventionally named origin).

Bash

Link your local repo to a remote GitHub URL

```
git remote add origin https://github.com/your-username/your-repo-name.git
```

Verify the remote connection

```
git remote -v
```

Push your local 'main' branch to GitHub for the first time

```
git push -u origin main
```

3. Daily Collaboration Commands

When working with a team, your daily workflow revolves around three core commands:

```
git push
```

Sends your local commits to the remote repository.

Bash

Push changes on your feature branch to GitHub

```
git push origin feature-login
```

git fetch

Downloads new commits and branches from GitHub *without* altering your local files. It lets you safely inspect what teammates have done.

Bash

git fetch origin

git pull

Downloads new commits from GitHub and automatically merges them into your current local branch (git fetch + git merge).

Bash

Always pull before starting new work to ensure you have the latest code

git pull origin main

4. The Pull Request (PR) & Code Review Process

In professional engineering teams, developers almost never push code directly to main. Instead, they use Pull Requests (PRs).

1. Create local branch --> 2. Push branch to GitHub --> 3. Open Pull Request

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5. Merge into main <-- 4. Team Code Review & Approval <-----

1. Push Feature Branch: Push your isolated work to GitHub (git push origin feature-xyz).
2. Open a PR: Go to GitHub and click "Compare & pull request". Describe what changes were made and why.
3. Code Review: Team members review the code, leave feedback, or request changes inline.
4. Merge: Once approved (and automated CI tests pass), the PR is merged into main directly on GitHub.
5. Clean Up: Pull the updated main branch back to your computer (git pull origin main).

Practical Exercise

Simulate a full collaborative workflow on GitHub:

1. Create a public repository on GitHub named github-practice.
2. Connect your local repository from Lesson 1 to it using git remote add origin <URL>.
3. Create a feature branch locally called add-readme-details, add a line to README.md, and commit it.
4. Push the branch to GitHub: git push origin add-readme-details.
5. Open your browser, navigate to the repo, create a Pull Request, click Merge Pull Request, and delete the remote branch.

6. Switch back to main on your terminal and run git pull origin main to sync your local setup!